Dohyun Kim

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Education

KAIST — Daejeon, South Korea

Master of Science in Mechanical Engineering — *Mar. 2022 – Mar. 2024*

- Conducted graduate study and research in stereo matching, 3D perception, and camera geometry for robotics-relevant scene understanding.
- Studied LiDAR, camera calibration, sensor geometry, and multi-view perception foundations in the Department of Mechanical Engineering.

KAIST — Daejeon, South Korea

Bachelor of Science in Mechanical Engineering; Double Major in Electrical Engineering — *Mar. 2015 – Jun. 2021*

Technical Skills

3D Vision: camera calibration, stereo matching, LiDAR, multi-view geometry, 3D reconstruction, sensor geometry
Neural 3D: NeRF, Gaussian Splatting, neural scene representation, capture/reconstruction workflows
Robotics: robot perception foundations, robot guidance interest, spatial representation, Physical AI interest
AI / Agents: AI agents, agentic workflows, tool-use workflows, model-assisted automation
Programming: Python TBD, C++ TBD, PyTorch TBD, OpenCV TBD, ROS TBD, CUDA TBD

Work and Experience

Recon Labs — Seoul, South Korea

AI Researcher — *Mar. 2024 – Present*

- **Neural 3D Representation**: Worked on 3D graphics and reconstruction-related workflows involving neural 3D representations such as NeRF and Gaussian Splatting.
- **Real-world Capture**: Built practical understanding of real-world 3D capture, reconstruction quality, sparse observations, and spatial representation constraints.
- **AI Agents**: Explored AI agent workflows that connect model reasoning with tool use, automation, and applied engineering processes.
- **Robotics Bridge**: Developed a bridge between classical sensor geometry and modern neural 3D representation for robotics perception applications.

HUMAN Laboratory, KAIST — Daejeon, South Korea

Graduate Researcher — *Mar. 2022 – Mar. 2024*

- **Stereo Matching**: Worked on master's research related to stereo matching and 3D perception, connecting image geometry with depth estimation and spatial scene understanding.
- **Sensor Geometry**: Studied LiDAR sensing, camera calibration, and multi-view geometry as technical foundations for robot perception and sensor-based autonomy.
- **Robotics Perception**: Developed a robotics-oriented understanding of how sensing constraints, calibration quality, and geometric assumptions affect 3D perception performance.

Publications

Conference Proceeding: Jeongmin Kim, **Dohyun Kim**, Jin Kim, and Yong-Hwa Park, "A novel stereoscopic 3D motion measurement using bi-axial optical system with one image sensor for remote machine health monitoring," *Proc. SPIE 12899, MOEMS and Miniaturized Systems XXIII*, 128990I, 2024. DOI: 10.1117/12.3002065.

Domestic Conference: **Dohyun Kim**, "Stereo matching technique for large-scale ship block using epipolar geometry," *KSPE 2023 Autumn Conference*, Nov. 17, 2023.